HW1 Sol.

Some of the solutions are taken from the solution manual of the book

6.3.14a

First, note that

$$Y_n = \frac{\sum_{i=1}^{n} X_i^2}{n}$$

has asymptotically the normal distribution with mean σ^2 and variance $2\sigma^4/n$. Here, we have used the fact that $E(X_i^2) = \sigma^2$ and $E(X_i^4) = 2\sigma^4$.

(a) Let $g(x) = 1/x$. Then $g'(x) = -1/x^2$. So, the asymptotic distribution of $g(Y_n)$ is the normal distribution with mean $1/\sigma^2$ and variance $(2\sigma^4/n)/\sigma^8 = 2/[n\sigma^4]$.

7.5.2

Exercise 2: It was shown in Example 7.5.4 that the M.L.E. is $\bar{x}_n$. In this exercise,

$$\bar{x}_n = \frac{58}{70} = \frac{29}{35}.$$

7.5.6

Let $\theta = \sigma^2$. Then the likelihood function is

$$f_n(\mathbf{x} \mid \theta) = \frac{1}{(2\pi\theta)^{n/2}} \exp\left\{-\frac{1}{2\theta}\sum_{i=1}^{n}(x_i - \mu)^2\right\}.$$

7.6.2

The standard deviation of the Poisson distribution with mean θ is $\sigma = \theta^{1/2}$. Therefore, $\hat{\sigma} = \hat{\theta}^{1/2}$. It was found in Exercise 5 of Sec. 7.5 that $\hat{\theta} = \bar{X}_n$.

7.6.6

The distribution of $Z = (X - \mu)/\sigma$ will be a standard normal distribution. Therefore,

$$0.95 = \Pr(X < \theta) = \Pr\left(Z < \frac{\theta - \mu}{\sigma}\right) = \Phi\left(\frac{\theta - \mu}{\sigma}\right).$$

Hence, from a table of the values of Φ, it is found that $(\theta - \mu)/\sigma = 1.645$. Since $\theta = \mu + 1.645\sigma$, it follows that $\hat{\theta} = \hat{\mu} + 1.645\hat{\sigma}$. By example 6.5.4, we have

$$\hat{\mu} = \overline{X}_n \quad \text{and} \quad \hat{\sigma} = \left[\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2\right]^{1/2}.$$

7.6.12

We know that $\hat{\beta} = 1/\bar{X}_n$. Also, since the mean of the exponential distribution is $\mu = 1/\beta$, it follows from the law of large numbers that $\bar{X}_n \xrightarrow{p} 1/\beta$. Hence, $\hat{\beta} \xrightarrow{p} \beta$.

7.6.18

The mean of a Bernoulli random variable with parameter p is p. Hence, the method of moments estimator is the sample mean, which is also the M.L.E.

7.6.20

The mean of a Poisson random variable is θ, hence the method of moments estimator of θ is the sample mean, which is also the M.L.E.

Lec 3 Ex 1.2

1. $g(Z_n) = (Z_n - 1)^2$

Take

$$g(z) = (z-1)^2.$$

Since $Z_n \to 0$ in probability, expand around 0.

We have

$$g(0) = 1, \qquad g'(z) = 2(z-1), \qquad g'(0) = -2.$$

Since $g'(0) \neq 0$, the ordinary delta method gives

$$\sqrt{n}\big(g(Z_n) - g(0)\big) \;=\; \sqrt{n}\big((Z_n-1)^2 - 1\big) \xrightarrow{d} g'(0)\, N(0,1).$$

Hence

$$\sqrt{n}\big((Z_n-1)^2 - 1\big) \xrightarrow{d} N(0,4).$$

2. $g(Z_n) = Z_n^2$

Now take

$$g(z) = z^2.$$

Again $g(0) = 0$, but now

$$g'(0) = 0.$$

What goes wrong with the usual delta method?

The first-order delta method would say

$$\sqrt{n}\big(g(Z_n) - g(0)\big) = \sqrt{n}\, Z_n^2 \xrightarrow{d} g'(0)\, N(0,1) = 0.$$

This is true, but it is degenerate, so it does **not** capture the real nontrivial asymptotic behavior. The issue is that the first derivative vanishes at the expansion point, so the first-order term disappears.

How to fix it

Use a **second-order delta method** (or simply square the limit).

Since

$$\sqrt{n}\, Z_n \xrightarrow{d} N(0,1),$$

we can write

$$nZ_n^2 = (\sqrt{n}\, Z_n)^2.$$

By the continuous mapping theorem,

$$nZ_n^2 \xrightarrow{d} \chi_1^2,$$

because if $X \sim N(0,1)$, then $X^2 \sim \chi_1^2$.

So the correct nondegenerate asymptotic distribution is

$$nZ_n^2 \xrightarrow{d} \chi_1^2.$$

Lec 4 Ex 2

Problem 2

Let $X_1, X_2, \ldots, X_n$ be i.i.d Ber(θ) where $0 \le \theta \le 1$. What is the MLE of θ?

$$\mathbf{Ber}(p) = p^x (1-p)^{1-x}$$

$$\sum_{i=1}^n \log p_\theta(x_i) = \sum_{i=1}^n x_i \log(p) + (1-x_i)\log(1-p)$$

$$= n\log(1-p) + \sum_{i=1}^n x_i \log(p) - \sum_{i=1}^n x_i \log(1-p)$$

$$\arg\max_p \sum_{i=1}^n \log p_\theta(x_i) = \frac{-n}{1-p} + \frac{1}{p}\sum_{i=1}^n x_i \log(p) + \frac{1}{1-p}\sum_{i=1}^n x_i = 0$$

$$\frac{1}{p}\sum_{i=1}^n x_i \log(p) + \frac{1}{1-p}\sum_{i=1}^n x_i = \frac{n}{1-p}$$

$$(1-p)\sum_{i=1}^n x_i + p\sum_{i=1}^n x_i = np$$

$$\sum_{i=1}^n x_i - p\sum_{i=1}^n x_i = pn - p\sum_{i=1}^n x_i$$

$$\hat{p}_{MLE} = \frac{1}{n}\sum_{i=1}^n x_i$$